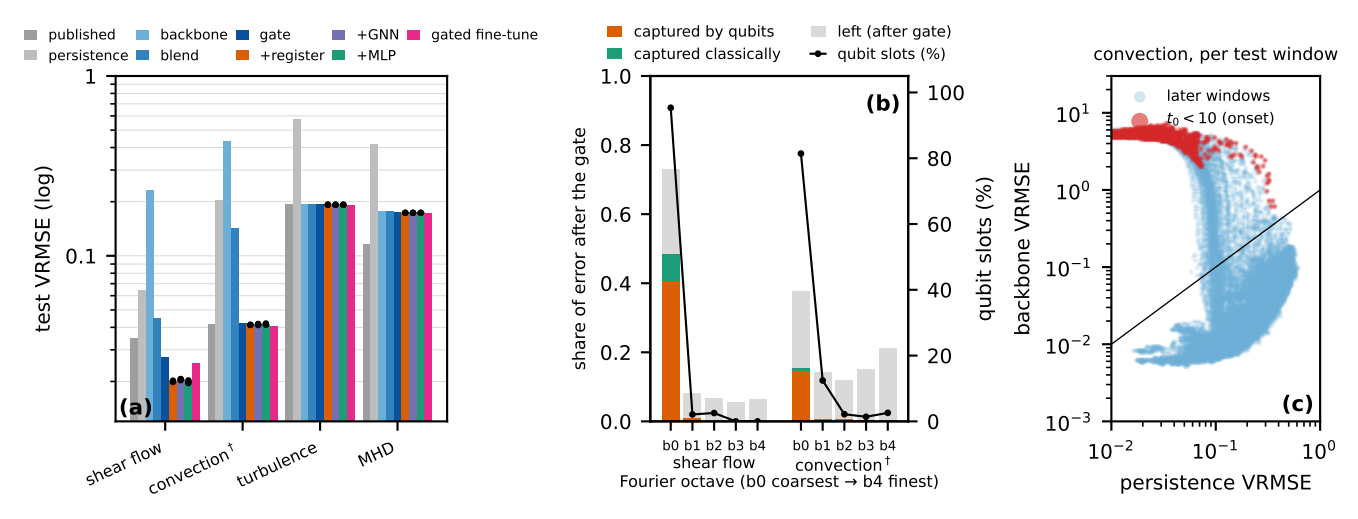
published
persistence
backbone
blend
gate
+register
+GNN
+MLP
gated fine-tune
test VRMSE (log)
1
0.1
(a)
shear flow
convection†
turbulence
MHD
captured by qubits
captured classically
left (after gate)
qubit slots (%)
share of error after the gate
1.0
0.8
0.6
0.4
0.2
0.0
100
80
60
40
20
0
(b)
b0 b1 b2 b3 b4
shear flow
convection†
Fourier octave (b0 coarsest → b4 finest)
convection, per test window
later windows
$t_0 < 10$ (onset)
backbone VRMSE
persistence VRMSE
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
(c)